

MA3236 NONLINEAR PROGRAMMING

Semester 1, AY 2026/2027

Tutorial 4: One-dimensional methods and Newton methods

Review: Lecture 8

- **Iteration and stopping.** On an open domain $\mathcal{D} \subseteq \mathbf{R}^n$, initialize, update, and test:

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha^{(k)} \mathbf{d}^{(k)} \in \mathcal{D}.$$

Typical tests use a small gradient norm, a small change in iterates or function values, or a resource limit. For example,

$$\|\nabla f(\mathbf{x}^{(k)})\| \leq \epsilon_g \quad \text{or} \quad \|\nabla f(\mathbf{x}^{(k)})\| \leq \epsilon_g \max\{1, \|\nabla f(\mathbf{x}^{(0)})\|\}.$$

A small step may signal stagnation; reaching the iteration limit is not a convergence certificate. If f^* is known, an objective-gap test is also possible.

- **Restricting to a line.** For a fixed direction, define

$$\phi^{(k)}(\alpha) = f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)}), \quad \mathcal{I}^{(k)} = \{\alpha \geq 0 : \mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)} \in \mathcal{D}\}.$$

Exact line search minimizes $\phi^{(k)}$ over $\mathcal{I}^{(k)}$, when the minimum is attained. Solving $(\phi^{(k)})' = 0$ alone need not find that minimum.

- **Rates and cost.** For $e^{(k)} = \|\mathbf{x}^{(k)} - \mathbf{x}^*\|$,

$$\begin{array}{l|l} \begin{array}{l} Q\text{-linear} \\ Q\text{-superlinear} \\ Q\text{-quadratic} \end{array} & \begin{array}{l} e^{(k+1)} \leq ce^{(k)} \text{ eventually, } 0 < c < 1 \\ e^{(k+1)}/e^{(k)} \rightarrow 0 \\ e^{(k+1)} \leq C(e^{(k)})^2 \text{ eventually.} \end{array} \end{array}$$

Iteration count, derivative evaluations, linear solves and storage are different costs. A local rate is not a guarantee from every initial point.

- **Three scalar methods.** Write $\ell^{(k)} = b^{(k)} - a^{(k)}$ and $\rho = (\sqrt{5} - 1)/2$.

Method	Assumption and update	Main cost / rate
Bisection on f'	Continuous f' and a sign-changing bracket; test its midpoint and retain the sign-changing half.	One new f' per reduction; $\ell^{(k+1)} = \ell^{(k)}/2$.
Newton on f'	$x^{(k+1)} = x^{(k)} - f'(x^{(k)})/f''(x^{(k)})$; the denominator must be nonzero.	One f' and f'' per update; locally Q-quadratic under regularity.
Golden section	Strict unimodality on a closed bracket; compare $f(\lambda^{(k)})$, $f(\mu^{(k)})$, where $\lambda^{(k)} = b^{(k)} - \rho\ell^{(k)}$, $\mu^{(k)} = a^{(k)} + \rho\ell^{(k)}$.	Two initial values, then one new value per comparison; $\ell^{(k+1)} = \rho\ell^{(k)}$.

For either bracketing method, the final midpoint error is at most $\ell^{(k)}/2$. This bound does not assert a fixed ratio of actual errors.

Review: Lecture 9

Set $\mathbf{g}^{(k)} = \nabla f(\mathbf{x}^{(k)})$ and $\mathbf{H}^{(k)} = \mathbf{H}_f(\mathbf{x}^{(k)})$.

- **Pure Newton: solve the model's stationarity equation.**

$$m^{(k)}(\mathbf{d}) = f(\mathbf{x}^{(k)}) + (\mathbf{g}^{(k)})^T \mathbf{d} + \frac{1}{2} \mathbf{d}^T \mathbf{H}^{(k)} \mathbf{d}, \quad \mathbf{H}^{(k)} \mathbf{d}_N^{(k)} = -\mathbf{g}^{(k)}.$$

Take the full step $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \mathbf{d}_N^{(k)}$. A nonsingular Hessian gives a unique solution of the equation, not necessarily a minimizer of the model or of f .

- **Local quadratic convergence.** At a stationary point $\mathbf{x}^*$, assume $\mathbf{H}_f(\mathbf{x}^*)$ is non-singular and $\mathbf{H}_f$ is locally Lipschitz. Pure Newton converges locally Q-quadratically when $\mathbf{x}^{(0)}$ is sufficiently close. Positive definiteness of $\mathbf{H}_f(\mathbf{x}^*)$ is not required: the target can be a saddle point.
- **Descent and shifting.** A descent direction satisfies $(\mathbf{g}^{(k)})^T \mathbf{d}^{(k)} < 0$ and decreases f for sufficiently small positive steps. Choose $\tau^{(k)} \geq 0$ so that

$$\mathbf{B}^{(k)} = \mathbf{H}^{(k)} + \tau^{(k)} I \succ 0, \quad \mathbf{B}^{(k)} \mathbf{d}_M^{(k)} = -\mathbf{g}^{(k)}.$$

Then $(\mathbf{g}^{(k)})^T \mathbf{d}_M^{(k)} < 0$ if $\mathbf{g}^{(k)} \neq \mathbf{0}$. The implementation tests, e.g., $\tau = 0, 0.1, 0.2, 0.4, \dots$ and uses the first positive-definite matrix, followed by a full step.

- **Positive-definite model $\neq$ majorization.** The modified model is

$$M^{(k)}(\mathbf{d}) = f(\mathbf{x}^{(k)}) + (\mathbf{g}^{(k)})^T \mathbf{d} + \frac{1}{2} \mathbf{d}^T \mathbf{B}^{(k)} \mathbf{d}.$$

If $M^{(k)}(\mathbf{d}) \geq f(\mathbf{x}^{(k)} + \mathbf{d})$ on the relevant region, minimizing it guarantees decrease. Positive definiteness alone does not imply this inequality: pure and shifted full steps can both increase f .

- **Armijo backtracking line search.** For a descent direction, start with $\alpha = 1$ and multiply it by $\beta \in (0, 1)$ until the trial point belongs to $\mathcal{D}$ and

$$f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)}) \leq f(\mathbf{x}^{(k)}) + \sigma \alpha (\mathbf{g}^{(k)})^T \mathbf{d}^{(k)}, \quad 0 < \sigma < 1.$$

Backtracking then terminates in finitely many trials. Newton–Armijo uses the raw Newton direction only if it is well defined and descending; otherwise it uses a shifted safeguard.

- **What line search does and does not promise.** It enforces sufficient decrease, not a universal global-minimizer guarantee. Pure Newton may converge without it. Near a regular local minimizer, the unit Newton step is eventually accepted when $0 < \sigma < 1/2$ and the local smoothness assumptions hold.

1. **Three one-dimensional methods.** Let $\varphi(t) = t^4 + 2t$ and $\psi(t) = t^2 + 2t$, both on $\mathcal{D} = \mathbf{R}$.

- (a) For φ , carry out two bisection reductions on φ' from $[-1, 1]$. How many reductions in total guarantee a midpoint error of at most 10^{-3} ?
- (b) For φ , compute two pure Newton updates from $t^{(0)} = -1$ exactly, without early stopping.
- (c) For ψ , carry out two golden-section reductions from $[-1, 1]$. Show the interior points, function-value comparisons and retained brackets, reusing a computed value when possible.

Solution. We have $\varphi'(t) = 4t^3 + 2$ and $\varphi''(t) = 12t^2$.

- (a) The derivative is strictly increasing and vanishes only at $t^* = -2^{-1/3}$; its sign shows that t^* is the unique global minimizer. The initial signs are $\varphi'(-1) = -2 < 0 < 6 = \varphi'(1)$.

k	$t^{(k)}$	$\varphi'(t^{(k)})$	$[a^{(k+1)}, b^{(k+1)}]$
0	0	2	$[-1, 0]$
1	$-1/2$	$3/2$	$[-1, -1/2]$.

After N reductions the width is 2^{1-N} , so the midpoint error is at most 2^{-N} . Hence $N = \lceil \log_2(1000) \rceil = 10$ suffices in total.

- (b) Newton's formula gives

$$t^{(1)} = -1 - \frac{-2}{12} = -\frac{5}{6}, \quad t^{(2)} = -\frac{5}{6} - \frac{-17/54}{25/3} = -\frac{179}{225} \approx -0.795556.$$

Here $\varphi'(-5/6) = -17/54$ and $\varphi''(-5/6) = 25/3$.

- (c) Write $\rho = (\sqrt{5} - 1)/2$, $\lambda^{(k)} = b^{(k)} - \rho(b^{(k)} - a^{(k)})$ and $\mu^{(k)} = a^{(k)} + \rho(b^{(k)} - a^{(k)})$. The function $\psi(t) = (t + 1)^2 - 1$ is strictly increasing on $[-1, 1]$, so it is strictly unimodal with endpoint minimizer -1 . For the first reduction,

$$\begin{aligned} \lambda^{(0)} &= 2 - \sqrt{5}, & \mu^{(0)} &= \sqrt{5} - 2, \\ \psi(\lambda^{(0)}) &= 13 - 6\sqrt{5} < 5 - 2\sqrt{5} = \psi(\mu^{(0)}), \\ [a^{(1)}, b^{(1)}] &= [-1, \sqrt{5} - 2]. \end{aligned}$$

For the second reduction, reuse $\mu^{(1)} = \lambda^{(0)}$:

$$\begin{aligned} \lambda^{(1)} &= 2\sqrt{5} - 5, & \mu^{(1)} &= 2 - \sqrt{5}, \\ \psi(\lambda^{(1)}) &= 35 - 16\sqrt{5} < 13 - 6\sqrt{5} = \psi(\mu^{(1)}), \\ [a^{(2)}, b^{(2)}] &= [-1, 2 - \sqrt{5}]. \end{aligned}$$

2. **Newton and exact line search on a quadratic.** Consider

$$q(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{Q}\mathbf{x} + \mathbf{c}^T \mathbf{x}, \quad \mathbf{Q} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathcal{D} = \mathbf{R}^2,$$

with $\mathbf{x}^{(0)} = (1, 1)^T$.

- (a) Compute the pure Newton direction and the next iterate. Why does this full step solve the minimization problem?
- (b) For a positive-definite quadratic and a descent direction $\mathbf{d}$, derive

$$\alpha^* = -\frac{\nabla q(\mathbf{x})^T \mathbf{d}}{\mathbf{d}^T \mathbf{Q} \mathbf{d}}$$

for exact line search over $\alpha \geq 0$. Apply it at $\mathbf{x}^{(0)}$ with $\mathbf{d}^{(0)} = (-1, -1)^T$: find the step and next point. Does this line search reach the global minimizer of q ?

Solution. The leading principal minors of $\mathbf{Q}$ are 2 and 5, so $\mathbf{Q} \succ 0$.

- (a) The gradient is $\nabla q(\mathbf{x}) = \mathbf{Q}\mathbf{x} + \mathbf{c}$, the Hessian is $\mathbf{Q}$, and $\mathbf{g}^{(0)} = (4, 5)^T$. Solving $\mathbf{Q}\mathbf{d}_N^{(0)} = -\mathbf{g}^{(0)}$ gives

$$\mathbf{d}_N^{(0)} = \begin{pmatrix} -7/5 \\ -6/5 \end{pmatrix}, \quad \mathbf{x}^{(1)} = \begin{pmatrix} -2/5 \\ -1/5 \end{pmatrix} = -\mathbf{Q}^{-1}\mathbf{c}.$$

The quadratic model is exact, and

$$q(\mathbf{x}) - q(\mathbf{x}^{(1)}) = \frac{1}{2}(\mathbf{x} - \mathbf{x}^{(1)})^T \mathbf{Q}(\mathbf{x} - \mathbf{x}^{(1)}) > 0 \quad (\mathbf{x} \neq \mathbf{x}^{(1)}).$$

Thus $\mathbf{x}^{(1)}$ is the unique global minimizer, with value $-3/10$.

- (b) Expansion gives

$$q(\mathbf{x} + \alpha \mathbf{d}) = q(\mathbf{x}) + \alpha \nabla q(\mathbf{x})^T \mathbf{d} + \frac{1}{2} \alpha^2 \mathbf{d}^T \mathbf{Q} \mathbf{d}.$$

A descent direction is nonzero, so $\mathbf{d}^T \mathbf{Q} \mathbf{d} > 0$. This scalar quadratic is strictly convex; setting its derivative to zero gives the stated $\alpha^* > 0$, which is the unique minimum over $\alpha \geq 0$. For the prescribed data,

$$(\mathbf{g}^{(0)})^T \mathbf{d}^{(0)} = -9, \quad (\mathbf{d}^{(0)})^T \mathbf{Q} \mathbf{d}^{(0)} = 7, \quad \alpha^* = \frac{9}{7}.$$

The next point is $(-2/7, -2/7)^T$, and

$$q(-2/7, -2/7) = -2/7 > -3/10.$$

Thus this exact line search does not reach the global minimizer of the two-dimensional problem: the chosen line does not pass through it.

3. **Shifting is not a step-length safeguard.** Use $\varphi(t) = t^4 + 2t$ from Question 1, now at $t^{(0)} = 0$.

- (a) Can pure Newton take a step? For shifted Newton, test $\tau = 0, 0.1, 0.2, 0.4, \dots$ and select the first positive scalar $B^{(0)} = \varphi''(t^{(0)}) + \tau$. Compute

$$d_M^{(0)} = -\frac{\varphi'(t^{(0)})}{B^{(0)}} \quad \text{and} \quad \varphi'(t^{(0)})d_M^{(0)}.$$

- (b) Compute the full shifted step and its function value. Does the positive-definite shifted model majorize φ at that trial point?
- (c) Apply Newton–Armijo with this shifted direction, $\bar{\alpha} = 1$, $\beta = 1/2$ and $\sigma = 10^{-4}$. List every trial step length up to acceptance, and give the accepted iterate. Explain which failure shifting fixes and which backtracking fixes.

Solution. At $t^{(0)} = 0$, $\varphi' = 2$ and $\varphi'' = 0$.

- (a) The pure Newton equation $0 \cdot d = -2$ has no solution. The first accepted shift is $\tau^{(0)} = 0.1$, so

$$B^{(0)} = 0.1, \quad d_M^{(0)} = -20, \quad \varphi'(0)d_M^{(0)} = -40 < 0.$$

- (b) The full step is $t^{(1)} = -20$, with $\varphi(-20) = 159960 > \varphi(0) = 0$. The shifted model is

$$M^{(0)}(d) = 2d + 0.05d^2, \quad M^{(0)}(-20) = -20.$$

Thus $M^{(0)}(-20) < \varphi(-20)$: the model is positive definite but does not majorize the objective at its minimizer.

- (c) At a trial step α , Armijo requires

$$160000\alpha^4 - 40\alpha \leq -0.004\alpha.$$

The successive tests are

α	$\varphi(-20\alpha)$	-0.004α	decision
1	159960	-0.004	reject
1/2	9980	-0.002	reject
1/4	615	-0.001	reject
1/8	34.0625	-0.0005	reject
1/16	-0.05859375	-0.00025	accept.

Hence $\alpha^{(0)} = 1/16$ and the Newton–Armijo iterate is $t^{(1)} = -5/4$. Shifting makes the singular equation solvable and supplies a descent direction, but does not restrict the full step. Backtracking makes the step short enough to enforce sufficient decrease.

4. **An exact Newton sequence and a stopping index.** Consider

$$f(x_1, x_2) = x_1^4 + 2x_1^2x_2^2 + x_2^4, \quad \mathcal{D} = \mathbf{R}^2,$$

with $\mathbf{x}^{(0)} = (a, a)^T$, where $a \neq 0$.

(a) Show that every full Newton update is well defined and

$$\mathbf{x}^{(k)} = \left(\frac{2}{3}\right)^k (a, a)^T.$$

(b) Identify the limit and the convergence rate of the iterates. Does this contradict the local quadratic convergence theorem?

(c) For $a = 1$, find the exact ratio $f(\mathbf{x}^{(k+1)})/f(\mathbf{x}^{(k)})$ and the smallest integer k for which $f(\mathbf{x}^{(k)}) \leq 10^{-6}$.

Solution. The gradient and Hessian are

$$\nabla f(\mathbf{x}) = 4\|\mathbf{x}\|^2\mathbf{x}, \quad \mathbf{H}_f(\mathbf{x}) = 4\|\mathbf{x}\|^2I + 8\mathbf{x}\mathbf{x}^T.$$

(a) At $(t, t)^T$ with $t \neq 0$,

$$\nabla f(t, t) = 8t^3 \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{H}_f(t, t) = 8t^2 \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \succ 0.$$

The Newton direction is $-(t/3)(1, 1)^T$, and the next iterate is $(2t/3, 2t/3)^T$. The formula holds at $k = 0$; substituting $t = (2/3)^k a \neq 0$ gives the formula at $k + 1$. Induction proves both the formula and nonsingularity at every finite iteration.

(b) Since $f(\mathbf{x}) = (x_1^2 + x_2^2)^2 \geq 0$ with equality only at $\mathbf{0}$, the limit $\mathbf{0}$ is the unique global minimizer. However,

$$\frac{\|\mathbf{x}^{(k+1)}\|}{\|\mathbf{x}^{(k)}\|} = \frac{2}{3},$$

so convergence is Q-linear, not Q-superlinear or Q-quadratic. The theorem does not apply because $\mathbf{H}_f(\mathbf{0}) = \mathbf{0}$ is singular. Nonsingularity at every finite iterate differs from nonsingularity at the limit.

(c) For $a = 1$,

$$f(\mathbf{x}^{(k)}) = 4 \left(\frac{16}{81}\right)^k, \quad \frac{f(\mathbf{x}^{(k+1)})}{f(\mathbf{x}^{(k)})} = \frac{16}{81}.$$

The least integer satisfying the tolerance is

$$k = \left\lceil \frac{\log(4 \times 10^6)}{\log(81/16)} \right\rceil = 10.$$

The values at $k = 9$ and $k = 10$ are approximately 1.8314×10^{-6} and 3.6175×10^{-7} , respectively.